

# From Definitions to Proofs

## An Overview Mathematical Writing

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# What is Mathematical Writing?

- Mathematical writing is the clear communication of mathematical ideas.
- It combines:
  - Precise language
  - Mathematical symbols
  - Equations
  - Logical reasoning
  - Examples and proofs
- Mathematical writing is used in:
  - Textbooks
  - Research papers
  - Assignments
  - Theses and dissertations
  - Lecture notes

# Definitions

- A definition gives a precise meaning to a mathematical term.
- Definitions remove ambiguity and make mathematical statements precise.
- Example:

## Definition

A prime number is a natural number greater than 1 having exactly two positive divisors: 1 and itself.

- Examples of prime numbers:

2, 3, 5, 7, 11

- A clear definition is the foundation for later theorems and proofs.

## Notation: Words vs Symbols

- Notation provides a concise way of representing mathematical ideas.
- Example in words:  
*Let  $x$  be a real number greater than zero.*

- Symbolically:

$$x \in \mathbb{R}, \quad x > 0$$

- Symbols make mathematical communication shorter and more precise.
- However, mathematical writing should use both words and symbols.

$$\forall x \in \mathbb{R}, \quad x^2 \geq 0$$

- Words explain the reasoning.
- Symbols express mathematical relationships efficiently.

# What is a Theorem?

- A theorem is a mathematical statement established to be true by a valid proof.
- A theorem generally has:
  - Assumptions or conditions
  - A mathematical statement
  - A conclusion

## Example Theorem

The sum of two even integers is even.

Let

$$a = 2m, \quad b = 2n,$$

where  $m, n \in \mathbb{Z}$ .

Then

$$a + b = 2m + 2n = 2(m + n).$$

- Since  $m + n$  is an integer,  $a + b$  is even.

# Proofs

- A proof is a logical argument that establishes why a mathematical statement must be true.
- A proof should:
  - Begin with clearly stated assumptions
  - Follow logical steps
  - Use appropriate mathematical reasoning
  - Reach a valid conclusion

## Example

Let  $a$  and  $b$  be even integers.

Since

$$a = 2m, \quad b = 2n,$$

we have

$$a + b = 2m + 2n = 2(m + n).$$

Since  $m + n \in \mathbb{Z}$ ,

$a + b$  is even.

# The Role of Examples

- Examples help the reader understand a mathematical statement.
- They provide intuition and demonstrate how an idea works.
- Example:

$$6 + 8 = 14$$

- This illustrates that the sum of two even numbers can be even.
- However, one example does not prove a general statement.

## Important Distinction

- **Example:** Helps us understand what can happen.
- **Proof:** Establishes why something must happen.

# Dos and Don'ts of Mathematical Writing

## Do

- Define notation before using it.
- Use complete sentences.
- Give appropriate examples.
- Explain logical steps.
- Display important equations.
- Be precise.
- Revise your writing.

## Don't

- Use unexplained symbols.
- Write disconnected equations.
- Treat examples as proofs.
- Skip important reasoning.
- Put everything in one paragraph.
- Use vague language.
- Assume the first draft is final.

# Conclusion

- Mathematical writing is more than writing equations.
- Definitions provide precision.
- Notation makes mathematical communication concise.
- Examples provide intuition and understanding.
- Theorems state mathematical facts.
- Proofs establish mathematical validity.
- Clear equations and logical organization improve readability.